Ulas Kamaci

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Summary

PhD in Electrical and Computer Engineering (UIUC) with 10 years of research experience in machine learning for computational imaging and inverse problems. Thesis develops deep learning reconstruction, diffusion models, and calibrated uncertainty quantification for a nonlinear imaging inverse problem. On NASA's ICON mission, built the image artifact-removal pipeline and data-quality flags used in the public data releases.

Education

University of Illinois Urbana-Champaign (UIUC) — Urbana, IL
PhD in Electrical and Computer Engineering — Defended Jul 2026
Thesis: *Inverse Theoretic Approaches for Slitless Spectral Imaging*
Advisor: Prof. Farzad Kamalabadi
Coursework: Inverse Problems and Learning, Machine Learning for Signal Processing, Pattern Recognition, Computer Vision, Computational Inference, Introduction to Optimization, Random Processes, Information Theory, Vector Space Signal Processing, Digital Signal Processing II, Digital Imaging, Optical Imaging

Middle East Technical University (METU) — Ankara, Turkey
BS in Electrical and Electronics Engineering — Jun 2017

Skills

Machine Learning	Generative Models (Diffusion Models, GANs), Bayesian Inference, Uncertainty Quantification (Deep Ensembles, Conformal Prediction), Transformers, U-Nets/CNNs, Optimization (ADMM/ISTA)
Core Tools	Python, PyTorch, NumPy, SciPy, Git, Linux
Also Used	Keras, Scikit-learn, Scikit-image, MATLAB, C/C++
Domains	Computational Imaging, Computer Vision, Signal Processing, Remote Sensing, Fourier Optics
Languages	Turkish (native), English (professional)

Experience

Graduate Researcher — Aug 2017 – Aug 2026
Coordinated Science Laboratory, University of Illinois Urbana-Champaign — *Urbana, IL*

PhD Thesis Research Python, PyTorch, Scikit-image — github.com/ukamaci/slitless

- Constructed a training dataset from ∼2,400 Hinode/EIS observations (2006–2025): per-pixel spectral fits as ground truth, simulated instrument measurements, quality masking, and leakage-free splits, yielding 23k training patches.
- Trained a **U-Net** that recovers intensity, velocity, and line-width maps from overlapping multi-order measurements, **halving** velocity error vs. the classical iterative solver at **60x** faster inference.
- Showed the U-Net with 2 detectors beats classical methods with 5, pointing to cheaper instrument designs.
- Developed a **conditional diffusion model** that matches the U-Net's accuracy while returning posterior samples; it cut velocity error **2.4–8.8x** vs. diffusion posterior sampling and sampled **27x** faster on the same GPU.
- Derived and implemented **Diffusion Posterior Sampling** with a DDPM prior for the nonlinear forward model.
- Built a **deep ensemble** of 8 conditional diffusion models for **2-12x** better-calibrated uncertainty quantification than any single model, and conformalized it for interval coverage guarantees.

ICON: NASA Explorer Mission PyTorch, Keras, NetCDF — github.com/UIUC-SINE/icon-fuv

- Owned the **Level 2.5** product of ICON's Far Ultraviolet imager, the nighttime ionospheric density retrieval, as its point of contact; added its quality flags and delivered each release to the mission's Science Data Center through 2025.
- Built the imager's **star and hot-pixel removal** for its calibrated (L1) data around a residual **U-Net** trained on 50k synthetic star-injected frames; used in the public data releases.
- Developed the L1 **uncertainty model**, estimating detector gain from flight data to separate photon shot noise from background noise; its 1-sigma errors weight the L2.5 retrieval and propagate into its published error bars.
- Served 3 months (2022, 2023) as ICON **Tohban**, the on-duty mission monitor: reviewed L1/L2 science data and spacecraft and instrument telemetry daily, **approved data for public release**, reported anomalies to instrument leads and the PI, and coordinated reprocessing with data product leads in weekly science operations calls.

MAS: NASA Small Spacecraft Technology Program Python, Keras, Scikit-image github.com/UIUC-SINE/MAS

- Optimized **sensor placement** for a high-resolution **diffractive imaging system** via subset selection, improving reconstruction **SSIM by 32%** over the conventional focal-plane design.
- Combined a **plug-and-play ADMM** solver with a learned CNN denoiser (DnCNN) for **multi-frame deconvolution**, improving reconstruction **PSNR by 2.4 dB** (SSIM 0.63 to 0.70) over a transform-learning baseline.

VISORS: NSF Space Weather Research Program Python, NumPy, Scikit-image github.com/UIUC-SINE/MAS

- Built an **imaging simulator** for formation-flying spacecraft with a diffractive lens, modeling the optical PSF with Fourier optics, sub-pixel shifts between frames, motion blur from linear drift, and detector read, dark-current, and Poisson noise.
- Designed an efficient **multi-frame image registration** algorithm and a plug-and-play ADMM reconstruction with a BM3D denoiser for highly noisy, motion-blurred measurements.

Research Mentoring

- Mentored undergraduate researcher Taowei Huang on deep learning for spectral-imaging inverse problems, including unrolled neural networks, distribution-shift analysis, and plug-and-play reconstruction (May 2024 – May 2025).
- Advised M.S. student Zifei Han on diffusion posterior sampling for astronomical imaging, from forward modeling through training and reconstruction experiments; co-authored the resulting BASP 2025 poster (Dec 2024 – Mar 2025).
- Guided undergraduate researcher Kaijie Li on machine learning for computational imaging and inverse problems, including forward modeling, reconstruction algorithms, and deep-learning methods (Aug – Nov 2025).

Teaching Assistant Jan 2021 – May 2026
Department of Electrical and Computer Engineering, UIUC *Urbana, IL*

- Delivered graduate-level lectures on deep networks, diffusion models, variational autoencoders, and optimization.
- Courses (7 semesters): ECE 544 Pattern Recognition (2x), ECE 558 Digital Imaging (2x), ECE 364 Programming Methods for Machine Learning, ECE 310 Digital Signal Processing, ECE 110 Introduction to Electronics.

Summer Intern Jul 2016 – Aug 2016
Aselsan Research Center *Ankara, Turkey*

- Reformulated patch-based **sparsifying transform learning** for **MRI** reconstruction as an ADMM algorithm and derived closed-form solutions to every subproblem.
- Coded the solver from scratch in MATLAB, improving PSNR over the original formulation at 5x undersampling.

Intern Sep 2015
Udea Electronics *Ankara, Turkey*

- Improved the accuracy of a Bluetooth-beacon fingerprinting **indoor positioning** system by optimizing beacon placement.

Intern Jul 2015
Mikro Tasarım *Ankara, Turkey*

- Designed a digital oscilloscope and a CORDIC-based waveform generator in **Verilog** on an **FPGA**; the oscilloscope measures max, min, mean, RMS, period, frequency, and rise/fall times of 12-bit sampled waveforms.

Selected Projects

- **GAN Data Augmentation for Brain Tumor Segmentation** (Computer Vision course, UIUC, Dec 2021; PyTorch). Augmented U-Net brain tumor **segmentation** (BraTS MRI) with **DCGAN**-synthesized image/mask pairs, reaching a **92.3** Dice score vs. 88.5 on real data alone.
- **FRC-GAN: Resolution-Aware Super-Resolution** (CS598PS, UIUC, Dec 2020; PyTorch). Designed a GAN loss based on **Fourier ring correlation**, a frequency-domain resolution measure from microscopy, for 4x **super-resolution**. Training SRGAN with it raised correlation with ground truth at every spatial frequency, removed SRGAN's fake high-frequency textures, and slightly raised SSIM (0.704 vs. 0.699).

Software

- **slitless-learned-recon** (github.com/ukamaci/slitless-learned-recon): Code and trained models for the ApJ 2026 paper, covering dataset construction, U-Net training, and reproduction scripts.
- **dps_slitless** (github.com/ukamaci/dps_slitless): Diffusion-model posterior sampling and uncertainty quantification for slitless spectral imaging.
- **slitless** (github.com/ukamaci/slitless): Simulation and reconstruction code for slitless spectral imaging.

- **icon-fuv** (github.com/UIUC-SINE/icon-fuv): Processing code for NASA ICON's Far Ultraviolet Imager, including star and hot-pixel removal, quality flagging, and uncertainty propagation.
- **MAS** (github.com/UIUC-SINE/MAS): Photon-sieve imaging simulation and reconstruction (sensor placement, plug-and-play ADMM, multi-frame registration), co-developed with E. Widloski.

Publications

- **Diffusion Posterior Sampling for Slitless Spectral Imaging** 2026
  U. Kamaci and F. Kamalabadi. *IEEE ICASSP*. [link]
- **Solar Slitless Spectroscopy with Learned Reconstruction** 2026
  U. Kamaci and F. Kamalabadi. *The Astrophysical Journal*. [link]
- **In-Flight Performance of the Far Ultraviolet Instrument (FUV) on ICON** 2023
  H. U. Frey, S. B. Mende, R. R. Meier, **U. Kamaci**, et al. *Space Science Reviews*. [link]
- **Optimal Measurement Configuration in Computational Diffractive Imaging** 2020
  E. Widloski, **U. Kamaci**, and F. Kamalabadi. *IEEE ICIP*. [link]
- **Compressive Photon-Sieve Spectral Imaging** 2018
  O. F. Kar, **U. Kamaci**, F. C. Akyon, and F. S. Oktem. *Computational Optical Sensing and Imaging (COSI), Optica*. [link]
- **Efficient Sparsity-Based Inversion for Photon-Sieve Spectral Imagers with Transform Learning** 2017
  U. Kamaci, F. C. Akyon, T. Alkanat, and F. S. Oktem. *IEEE GlobalSIP*. [link]
- **A Bayesian Framework for Accurate Determination of the Nighttime Ionospheric Parameters from the ICON FUV Observations** 2024
  H. Liu, J. Qin, **U. Kamaci**, and F. Kamalabadi. *Space Science Reviews*. [link]
- **Update of ICON-FUV hmF2 and NmF2 Comparison with External Radio Observations** 2023
  G. Wautelet, B. Hubert, J.-C. Gérard, T. J. Immel, H. U. Frey, F. Kamalabadi, **U. Kamaci**, and S. L. England. *Space Science Reviews*. [link]
- **Topside Plasma Flows in the Equatorial Ionosphere and Their Relationships to F-Region Winds near 250 km** 2022
  R. A. Heelis, Y.-J. Chen, M. D. Depew, B. J. Harding, T. J. Immel, Y.-J. Wu, S. B. Mende, J. D. Huba, C. R. Englert, A. W. Stephan, **U. Kamaci**, and F. Kamalabadi. *JGR Space Physics*. [link]
- **Conjugate Photoelectron Energy Spectra Derived from Coincident FUV and Radio Measurements** 2021
  J. M. Urco, F. Kamalabadi, **U. Kamaci**, B. J. Harding, H. U. Frey, S. B. Mende, J. D. Huba, S. L. England, and T. J. Immel. *Geophysical Research Letters*. [link]
- **First ICON-FUV Nighttime NmF2 and hmF2 Comparison to Ground and Space-Based Measurements** 2021
  G. Wautelet, B. Hubert, J.-C. Gérard, T. J. Immel, H. U. Frey, S. B. Mende, F. Kamalabadi, **U. Kamaci**, and S. L. England. *JGR Space Physics*. [link]

Poster Presentations

- **Toward Attaining the Diffraction Limit and Beyond in Astronomical Imaging** 2025
  F. Kamalabadi, **U. Kamaci**, and Z. Han. *BASP Frontiers Workshop*. [link]
- **Posterior Sampling for Solar Snapshot Spectroscopy with Score-Based Priors** 2025
  U. Kamaci and F. Kamalabadi. *BASP Frontiers Workshop*. [link]
- **Instantaneous Spectral Imaging with Learned Reconstruction** 2023
  U. Kamaci and F. Kamalabadi. *BASP Frontiers Workshop*. [link]
- **Sparsifying Transform Learning for Computational Photon-Sieve Spectral Imaging** 2017
  U. Kamaci, F. C. Akyon, T. Alkanat, and F. S. Oktem. *METU EE STAR Undergraduate Research Competition*.
- **Compressed Sensing in Optical Imaging Systems and Its Applications in Spectral Imaging** 2016
  METU Engineering Day Poster Competition.

Honors

- **3rd Place**, METU EE STAR Undergraduate Research Competition (of 32 finalists), for sparsifying transform learning in photon-sieve spectral imaging Mar 2017
- **Best Term Project**, EE230 Probability and Random Variables, METU, for maximum-likelihood estimation of star brightness under Poisson photon noise Jun 2015
- **Dean's High Honor List** (8 semesters), METU 2013 – 2017